

Tube Model Predictive Control

Qwen3.8-Flash-Next

A nominal MPC plan is computed for the disturbance-free model, yet the plant is disturbed at every step. This note asks six questions about turning such a plan into a guarantee: why a nominal plan says nothing about the real trajectory, what object keeps the deviation from drifting, how that object is folded into the constraints, why feasibility survives from one step to the next, what sets the size of the resulting tube and what a larger tube costs, and what changes when the tube shape is chosen online. The running example is a discrete-time double integrator with a bounded additive disturbance.

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1. Why does a plan that ignores the disturbance fail?

Question: The controller can measure x_k and knows the bound on w_k , so it can run MPC on the nominal model. Predict whether satisfying the state and input constraints along the nominal plan guarantees that the real trajectory also satisfies them. What must be bounded for the guarantee to hold, and does the disturbance bound alone bound it?

Running example. A discrete-time double integrator with a bounded additive disturbance,

$$x_{k+1} = Ax_k + Bu_k + w_k, \quad A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0.5 \\ 1 \end{pmatrix}, \quad w_k \in \mathcal{W}, \quad (1)$$

with state $x_k = \begin{pmatrix} p_k \\ v_k \end{pmatrix} \in \mathbb{R}^2$, input $u_k \in \mathbb{R}$, and disturbance set $\mathcal{W} = \{w : \|w\|_\infty \leq 0.1\}$. The state and input constraints are

$$\mathcal{X} = \{x : \|x\|_\infty \leq 5\}, \quad \mathcal{U} = \{u : |u| \leq 1\}. \quad (2)$$

Facts to reason with.

- The nominal model MPC plans on is $z_{k+1} = Az_k + Bv_k$, and the plan starts from $z_0 = x_0$.
- A has a repeated eigenvalue at 1: it is not Schur, and A^i does not decay as i grows.
- w_k satisfies the bound in Equation 1 at every step; nothing else about it is known.
- The nominal plan $(z_0, v_0, z_1, v_1, \dots)$ satisfies $z_i \in \mathcal{X}$ and $v_i \in \mathcal{U}$.

2. What keeps the deviation from drifting?

Question: Feed the controller a correction proportional to the measured deviation: fix a gain K and apply $u_k = v_k + K(x_k - z_k)$. The deviation then obeys its own recursion. What condition must a bounded set $\mathcal{Z}$ satisfy so that $e_k \in \mathcal{Z}$ implies $e_{k+1} \in \mathcal{Z}$ for every admissible w_k ? And why does $A + BK$ being Schur not, by itself, keep e_k bounded as k grows?

Setup. Let $e_k = x_k - z_k$ and choose $K = \begin{pmatrix} -0.5 & -1.2 \end{pmatrix}$, so that $A + BK$ is Schur. The control law $u_k = v_k + Ke_k$ turns the deviation into

$$e_{k+1} = (A + BK)e_k + w_k, \quad e_0 = 0. \quad (3)$$

Facts to reason with.

- $\mathcal{W}$ is compact, convex, and contains the origin.
- For sets $\mathcal{P}$ and $\mathcal{Q}$, write $\mathcal{P} \oplus \mathcal{Q} = \{p + q : p \in \mathcal{P}, q \in \mathcal{Q}\}$.
- $A + BK$ is Schur: every eigenvalue of $A + BK$ lies strictly inside the unit disk.
- The disturbance acts at every step, not just once.

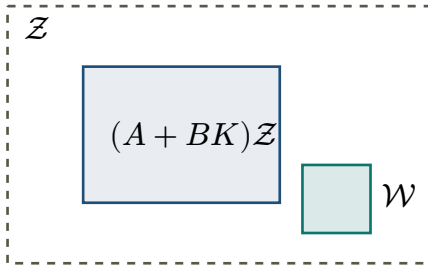


Figure 1: A candidate deviation set $\mathcal{Z}$ (dashed) and, inside it, the image $(A + BK)\mathcal{Z}$ together with the disturbance set $\mathcal{W}$. Inspect what has to be contained in what for the deviation to still lie in $\mathcal{Z}$ one step later.

3. How is the tube folded into the constraints?

Question: Once $x_k = z_k + e_k$ with $e_k \in \mathcal{Z}$, the state and input constraints must hold for the real trajectory, but the optimizer chooses only the nominal z_k and v_k . What must z_k and v_k satisfy so that $x_k \in \mathcal{X}$ and $u_k \in \mathcal{U}$ hold for **every** $e_k \in \mathcal{Z}$? And why can a single scalar margin on $\mathcal{X}$ and $\mathcal{U}$ not replace the same $\mathcal{Z}$ everywhere?

Setup. The tube is the family of sets $z_k \oplus \mathcal{Z}$ swept along the nominal plan, so the real state satisfies $x_k \in z_k \oplus \mathcal{Z}$. The applied input is $u_k = v_k + Ke_k$ with $e_k \in \mathcal{Z}$.

Facts to reason with.

- $\mathcal{Z}$ is a bounded set containing the origin; for the running example it is a polytope, not a disk, so its image $K\mathcal{Z} = \{Ke : e \in \mathcal{Z}\}$ is an interval whose width is set by the corners of $\mathcal{Z}$.
- $\mathcal{X}$ and $\mathcal{U}$ are the sets in Equation 2.
- The optimizer chooses z_k and v_k ; e_k is whatever the disturbance produces.
- The condition on z_k alone must hold simultaneously for all $e_k \in \mathcal{Z}$.

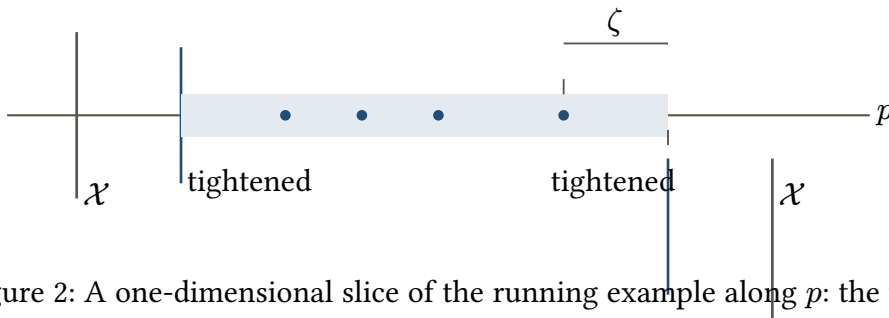


Figure 2: A one-dimensional slice of the running example along p : the tube of half-width ζ around each nominal point, the state-constraint boundary, and the tightened boundary.

Inspect which set the nominal points must stay inside.

4. Why does feasibility survive the disturbance?

Question: At each step the controller solves the nominal problem on the tightened constraints and applies the first input plus the correction. Take the plan that was feasible at time k and shift it forward by one step. Which shifted sequence is a candidate at time $k + 1$, and what must the terminal set $\mathcal{X}_f$ satisfy for that candidate to be feasible? Why does the disturbance not break the argument?

Setup. The nominal problem over horizon N uses the tightened constraints $z_i \in \mathcal{X}_{\text{tight}}$ and $v_i \in \mathcal{U}_{\text{tight}}$, and the terminal constraint $z_N \in \mathcal{X}_f$. At time k it returns $(z_{0|k}, v_{0|k}, \dots, z_{N|k})$, and the controller applies $u_k = v_{0|k} + K(x_k - z_{0|k})$.

Facts to reason with.

- The nominal successor of $z_{0|k}$ is $z_{1|k} = Az_{0|k} + Bv_{0|k}$.
- The error satisfies $e_1 = x_1 - z_{1|k} = (A + BK)e_0 + w_0$, with $e_0 \in \mathcal{Z}$.
- A linear terminal control law $u = K_f z$ is available for extending a plan past step N .
- The shifted plan already supplies N nominal points; only the point after the last one needs a tail.

5. What sets the size of the tube, and what does a larger tube cost?

Question: The gain K and the set $\mathcal{Z}$ are fixed before any optimization runs. How does the choice of K change the size of $\mathcal{Z}$, and which two quantities move in opposite directions as the tube grows? Why is it not enough to simply take the smallest set that satisfies the condition of the previous question, and what is paid for replacing it by a simpler shape?

Setup. Keep the running example. The tube $\mathcal{Z}$ satisfies $(A + BK)\mathcal{Z} \oplus \mathcal{W} \subseteq \mathcal{Z}$, and the tightened constraints are what the optimizer sees.

Facts to reason with.

- K enters both the closed-loop matrix $A + BK$ and the input tightening through $K\mathcal{Z}$.
- For any Schur $A + BK$ and bounded $\mathcal{W}$ there is a smallest closed set satisfying the condition, and it is generally not a disk.
- A polytope with few facets, or an ellipsoid, can be used as an outer approximation of that set.
- $\mathcal{X}_{\text{tight}}$ shrinks as $\mathcal{Z}$ grows, and $\mathcal{U}_{\text{tight}}$ shrinks as $K\mathcal{Z}$ grows.

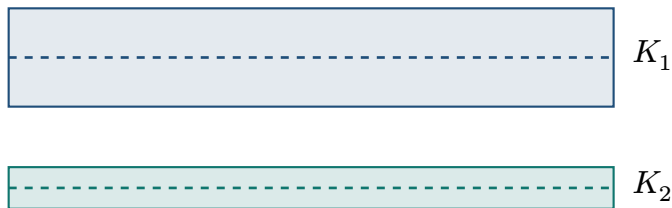


Figure 3: The same disturbance set $\mathcal{W}$ handled with two ancillary gains K_1 and K_2 . Inspect how the width of each tube and the range of the ancillary control Ke change together.

6. What changes when the tube shape is chosen online?

Question: So far K and $\mathcal{Z}$ are fixed offline and only the nominal plan is optimized. A less conservative design lets the tube itself vary, for instance $x_k \in z_k \oplus \alpha_k \mathcal{Z}$ with a fixed shape $\mathcal{Z}$ and scalars $\alpha_k \geq 0$ chosen online. What must be added to the optimization to keep the invariance condition true at every step, and what makes the more general choice of a free set $\mathcal{Z}_k$ hard to keep convex?

Setup. Let the tube at step k be $z_k \oplus \alpha_k \mathcal{Z}$ with $\alpha_k \geq 0$ chosen online, and apply the tightening of the previous question with $\alpha_k \mathcal{Z}$ in place of $\mathcal{Z}$. The invariance requirement must hold along the horizon.

Facts to reason with.

- For a compact set $\mathcal{S}$, its support function is $h_{\mathcal{S}}(d) = \max_{s \in \mathcal{S}} d^T s$.
- $\mathcal{P} \oplus \mathcal{Q}$ has $h_{\mathcal{P} \oplus \mathcal{Q}}(d) = h_{\mathcal{P}}(d) + h_{\mathcal{Q}}(d)$, and $\mathcal{P} \subseteq \mathcal{Q}$ exactly when $h_{\mathcal{P}}(d) \leq h_{\mathcal{Q}}(d)$ for every direction d .
- A common design keeps the shape $\mathcal{Z}$ fixed and scales it, so the tube is determined by the scalars α_k .
- In the general design each $\mathcal{Z}_k$ is itself a decision variable.

7. Sources and further reading

No external sources have been consulted yet; this note makes no external claims.